WEB PLATFORM

Real-time Web Visual Guide

Polling, SSE, and WebSockets — a decision guide with protocol flows for live data.

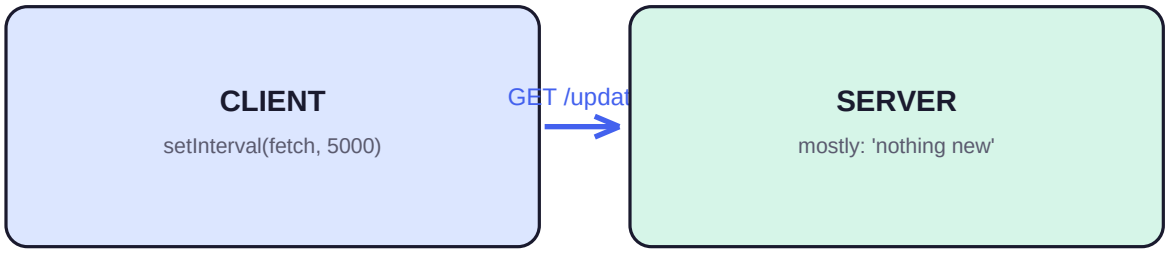
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Polling: Ask Repeatedly

Client fires requests on a timer. Simple, wasteful, laggy.



Latency up to the interval. 12 req/min/client even when idle — multiply by users.

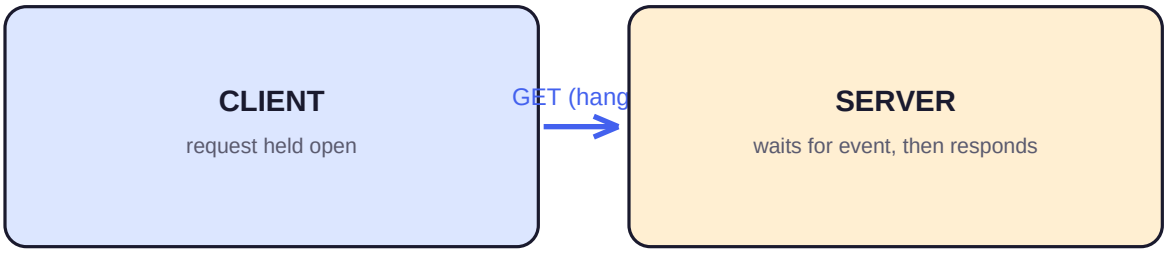
USE WHEN

rarely-changing data, tiny scale, zero infra appetite

STEP 2/6

Long Polling: Ask and Wait

Server holds the request open until data arrives or timeout.



Near-instant delivery, but each event costs a new connection + held server threads.

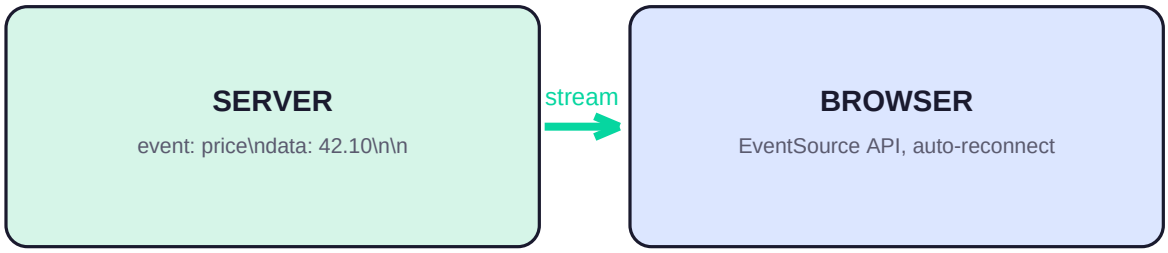
USE WHEN

server->client only, moderate scale, no socket infra

STEP 3/6

Server-Sent Events: One-Way Stream

A long-lived HTTP response; the server pushes text/event-stream chunks.



STRENGTHS

HTTP-friendly (proxies, auth), automatic reconnection, simple

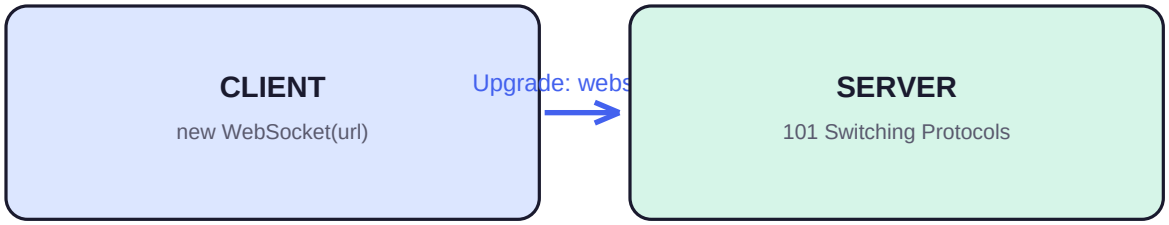
LIMIT

server -> client only; ~6 connections per browser

STEP 4/6

WebSockets: Full-Duplex Socket

HTTP upgrades to a persistent bidirectional frame channel.



persistent frame channel — either side sends anytime

USE WHEN

chat, games, collaborative editing — true bidirectional, low latency

Needs sticky sessions or shared pub/sub behind the LB; heartbeats detect dead peers.

STEP 5/6

Decision Guide

Match the pattern to the traffic shape.

Polling	5s+ staleness OK scale: tiny
Long polling	server->client, instant scale: medium
SSE	feeds, tickers, logs scale: large 1-way
WebSockets	bidirectional, <100ms scale: large 2-way

STEP 6/6

Scaling Real-time

Connections are stateful — plan for it.

Connection fan-out	one origin can't hold 1M sockets — shard across nodes
Pub/sub backbone	Redis/NATS/Kafka broadcast events to all socket servers
Sticky or stateless	route by user id, or keep zero per-connection state
Backpressure	slow consumers get sampled/coalesced updates, not an unbounde

STEP ✓/✓

Key Takeaways

Real-time Web — Remember

- Polling is simplest but wastes requests and adds interval latency.
- Long polling gives near-instant server->client delivery at connection cost.
- SSE streams server->client over plain HTTP with auto-reconnect — great for feeds.
- WebSockets are full-duplex and lowest-latency; use for chat, games, collaboration.
- Decide by direction, latency budget, and scale — not by hype.
- Scale with sharded socket servers, a pub/sub backbone, and backpressure for slow clients.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev